ERHVERVSAKADEMI
AARHUS

Machine Learning – Week 2

Welcome - Todays Program.

- Python Exercises
- Up and running with Jupyter Notebooks.
- Linear Regression.

Python Exercises.

Tal sammen i grupper af 2.

- Hvordan gik det med de opgaver?
- Python som sprog, hvad synes indtil videre?

Jupyter – Up and Running.

Se filen JupyterUpAndRunnng.pdf

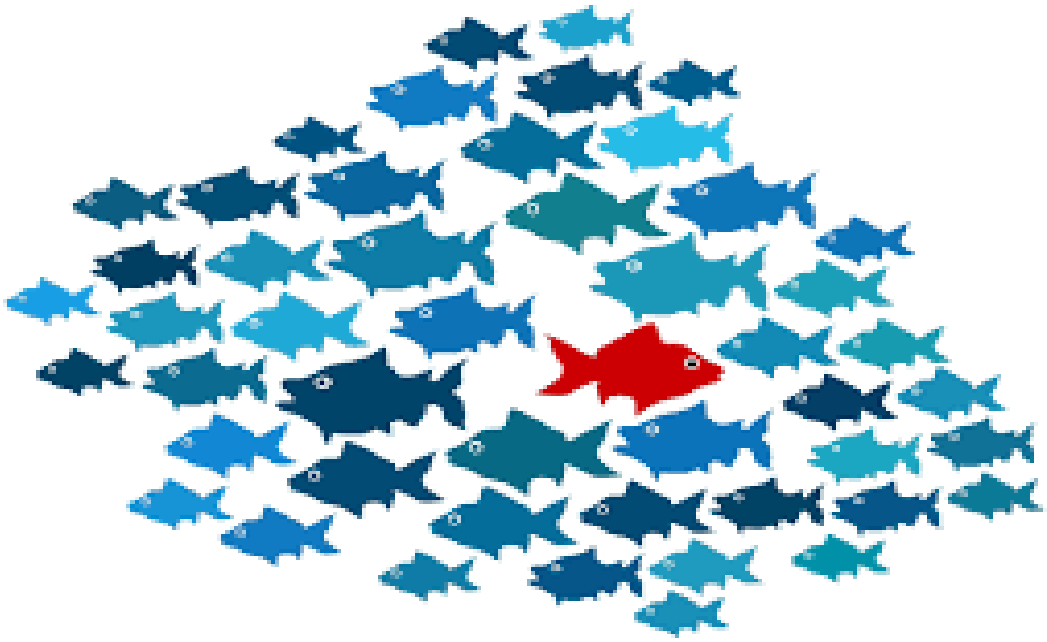
- Vi skal have en Jupyter Notebook up and running.

- Følg guiden RunningJupyter_QuickGuide.pdf, og tal sammen i grupper af 2.

(Når alle lige har fået lidt tid til at sætte sig ind i problemstillingen kommer der også lidt hjælp på projektoren).

What can Machine Learning do?

Detect anomalies (security, fraud).



What can Machine Learning do?

Healthcare – Second opinion:

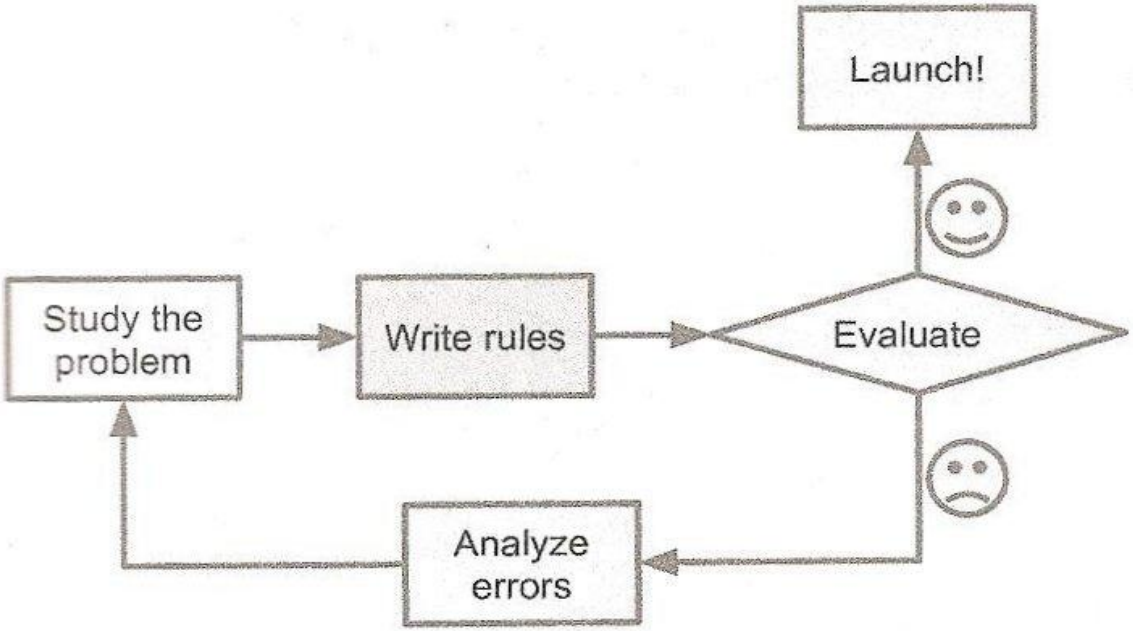
- Historical datasets of patient characteristics.
- Data and algorithm are combined to create predictive model.
- Predictive models assigns probabilities.
- Patients outcome is predicted.

What can Machine Learning do?

Customer segmentation
for targeted marketing.

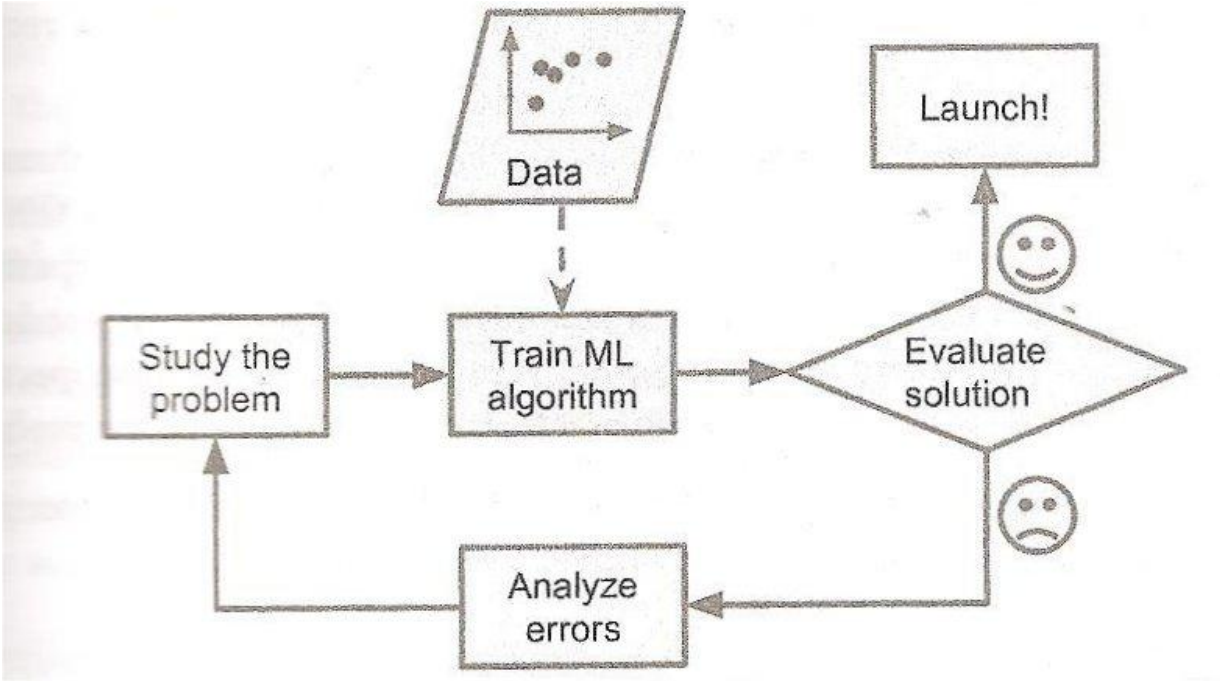


What can Machine Learning do?



Traditional approach.
p. 5 v2. "Hands-On ML"
by Aurelien Geron.

What can Machine Learning do?



ML approach.

p. 5 v2 "Hands-On ML"

by Aurelien Geron.

Machine Learning – use when:

Machine learning is great for:

- Problems for which existing solutions require a lot of hand tuning, or long lists of rules.
- Complex problems for which there is no good solution at all using traditional approach.
- Fluctuating environments: A machine learning system can adapt to new data.
- Getting insights about complex problems and large amounts of data.

p. 7. v2. ”Hands-On Machine Learning” by Aurelien Geron.

Machine Learning – How

Machine learning – How:

- You studied the data.
- You selected a model.
- You trained on the training data.
- Finally, you applied the model to make predictions on new cases.

p. 22. v2. ”Hands-On Machine Learning” by Aurelien Geron.

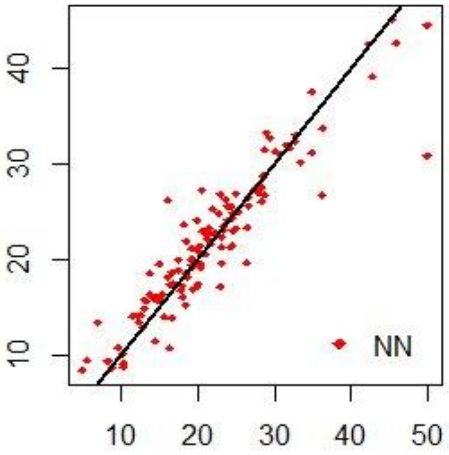
Machine Learning – How

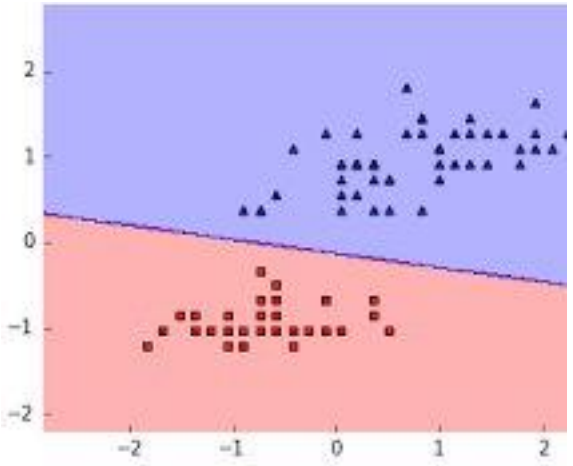
A machine learning project may not be linear, but it has a number of well known steps:

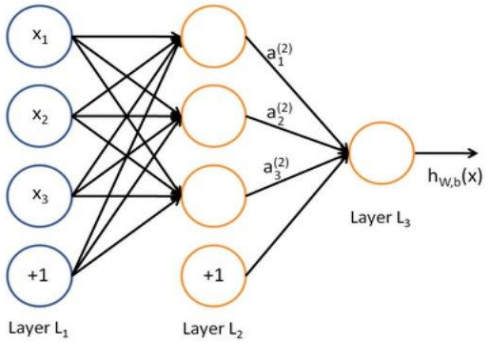
1. Define Problem.
2. Prepare Data.
3. Evaluate Algorithms.
4. Improve Results.
5. Present Results.

Machine Learning – Misc. Techniques.

Linear regression, logistic regression, neural nets and more.







Machine Learning – Big picture.

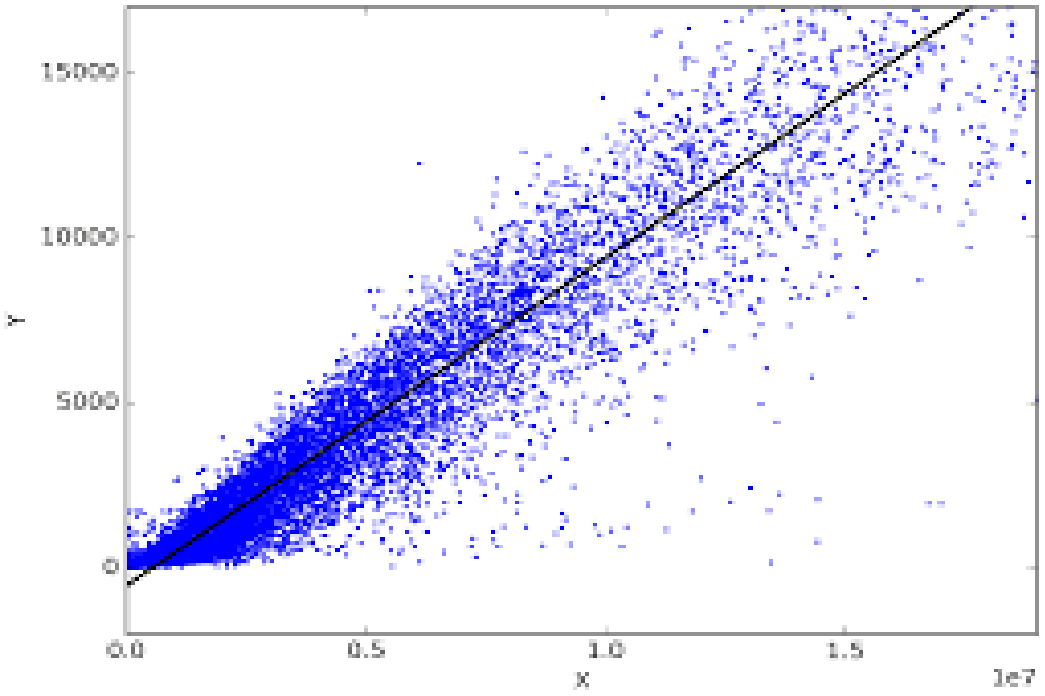
In your own words.

In groups of 2, discuss:

- How would you define Machine Learning?
- Can you name problems where it shines?
  And explain why it shines there?

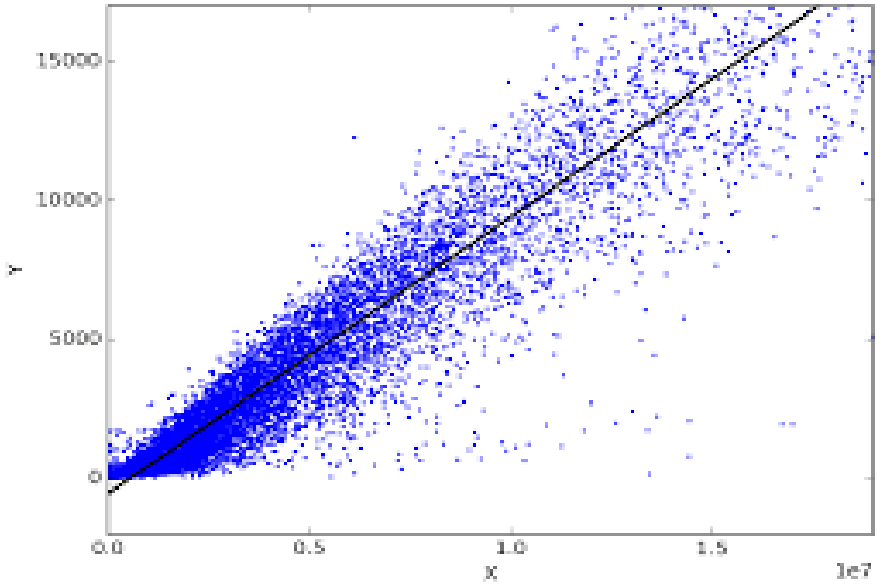
Machine Learning - Linear regression

Using LR:



Machine Learning – A plot using chatGPT

Using LR:



Ask chatGPT to help you come up Python code that
creates a line through the points, in a case where linear regression is possible.
See the exercise: LinearRegressionWithChatGPT

Machine Learning - Linear regression

Linear regression:

Hypothesis:

$$h_\theta(x) = \theta_0 + \theta_1 x$$

Parameters:

$$\theta_0, \theta_1$$

Cost Function:

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

Goal: $\underset{\theta_0, \theta_1}{\text{minimize}}\ J(\theta_0, \theta_1)$

Machine Learning - Linear regression

Linear regression, exact equations.

$$a = \frac{(\sum y)(\sum x^2) - (\sum x)(\sum xy)}{n(\sum x^2) - (\sum x)^2}$$

$$b = \frac{n(\sum xy) - (\sum x)(\sum y)}{n(\sum x^2) - (\sum x)^2}$$

Equation 4-4. Normal Equation

$$\hat{\theta} = \left(\mathbf{X}^T \cdot \mathbf{X}\right)^{-1} \cdot \mathbf{X}^T \cdot \mathbf{y}$$

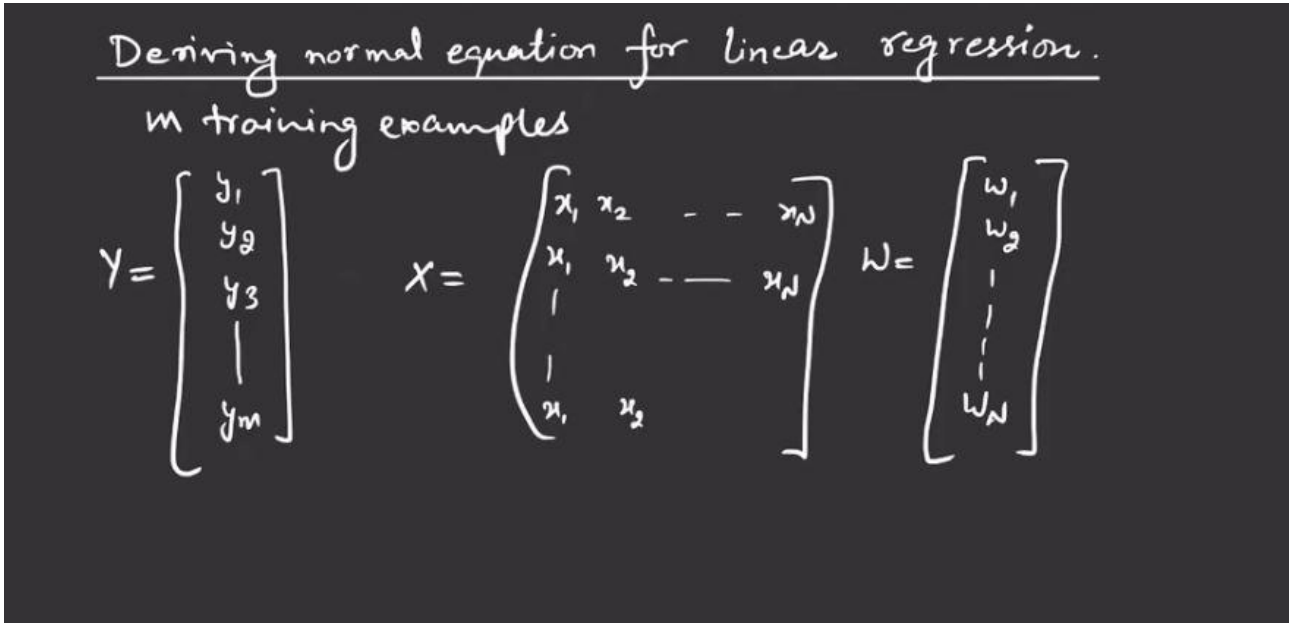
More about Linear regression on p. 108 v2. in "Hands-On ML" by Aurelien Geron. Equation 4.4. p 116 v2. "Normal Equation".

Machine Learning – Linear regression

Linear regression, exact equations.

Equation 4-4. Normal Equation

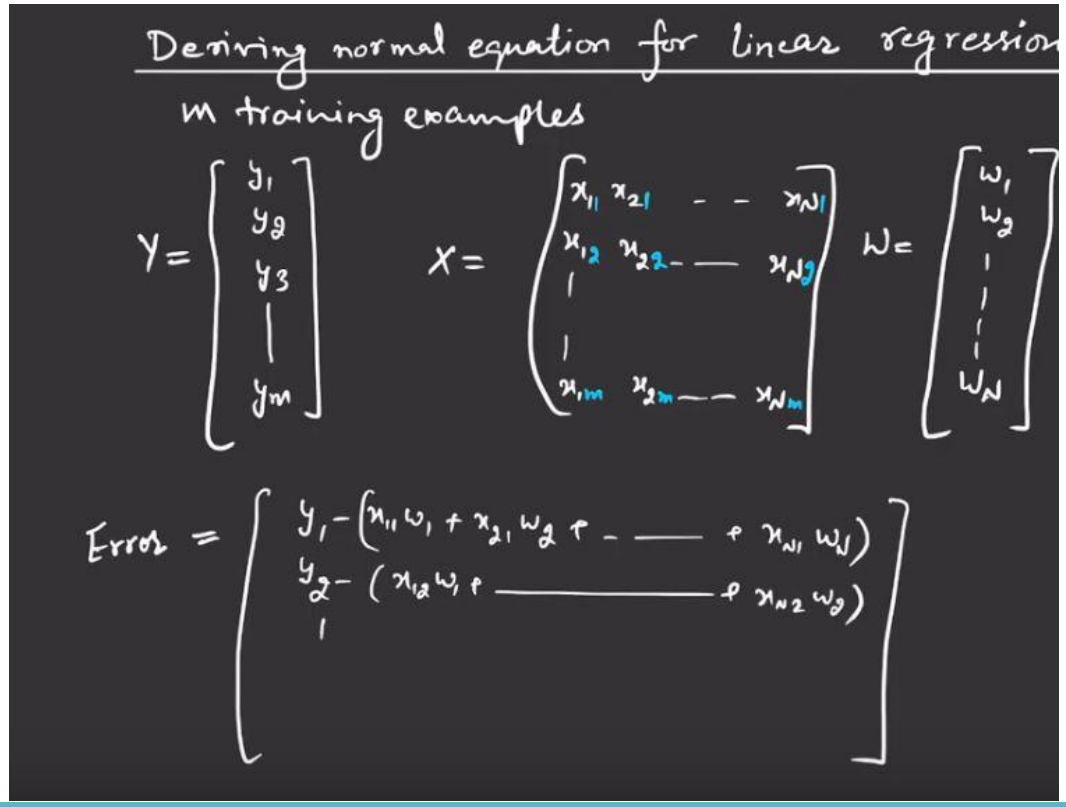
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Udledning af formel.
- Ikke en del af pensum -

Machine Learning – Linear regression

Linear regression, exact equations.



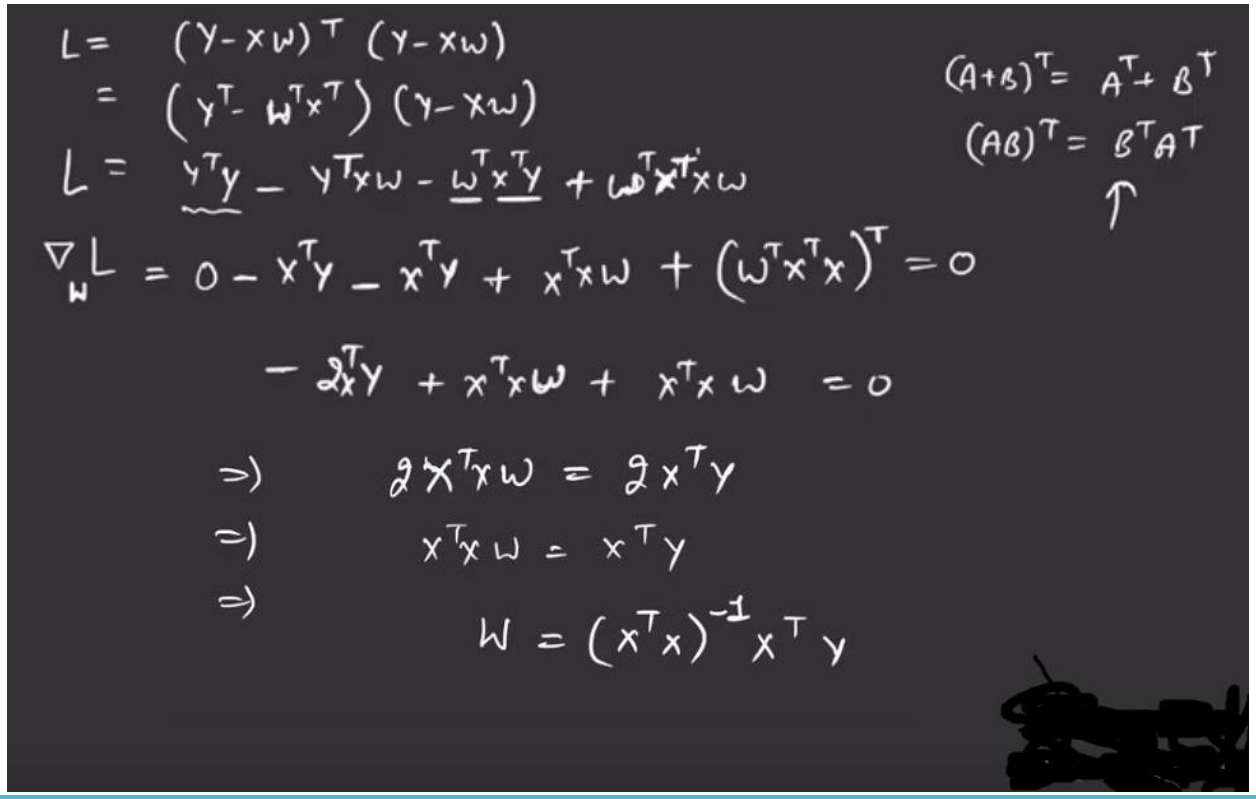
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Machine Learning – Linear regression

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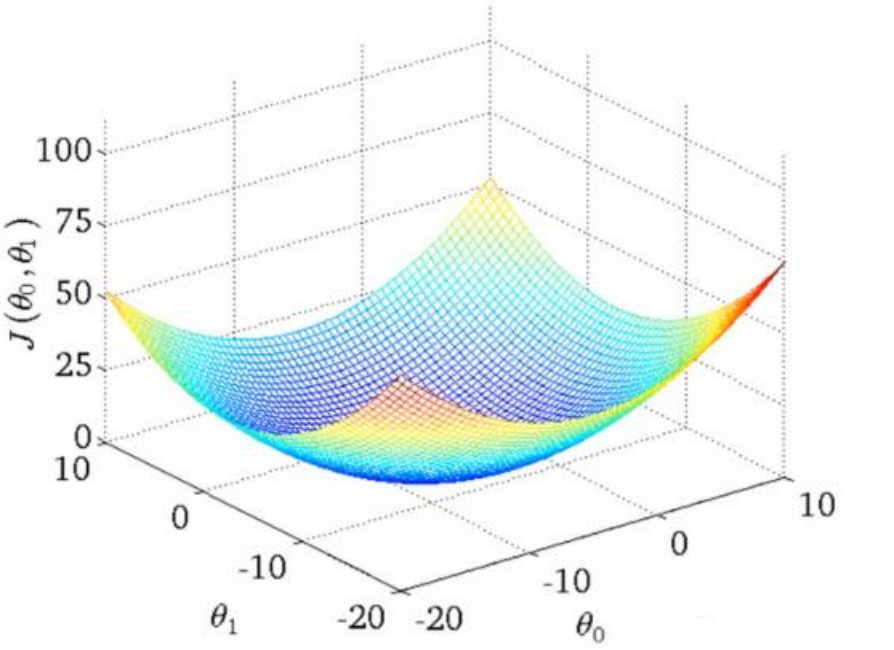
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Machine Learning - Linear regression

Linear regression, minimize cost function.



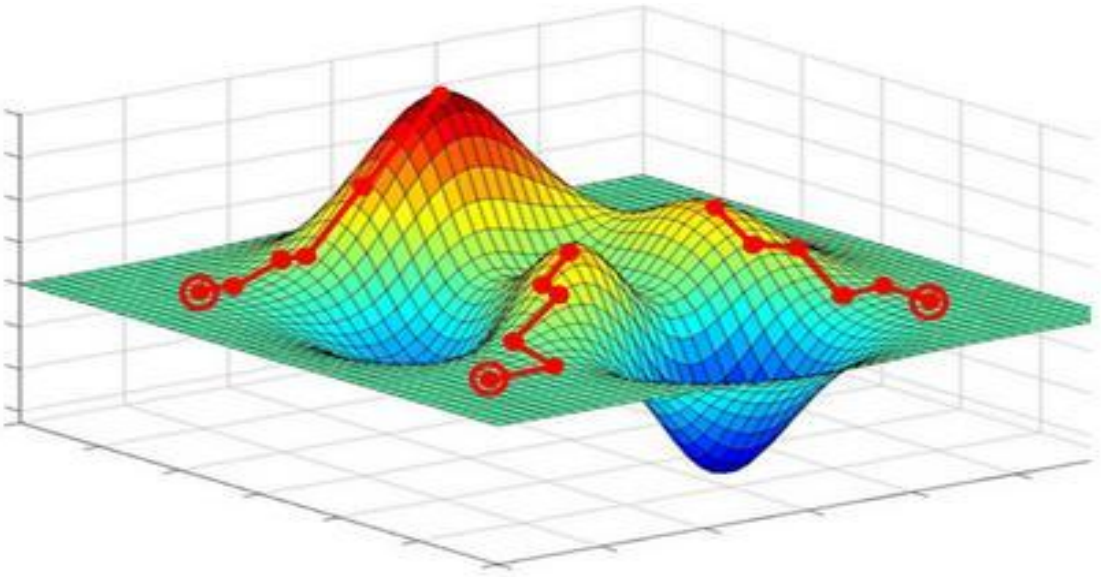
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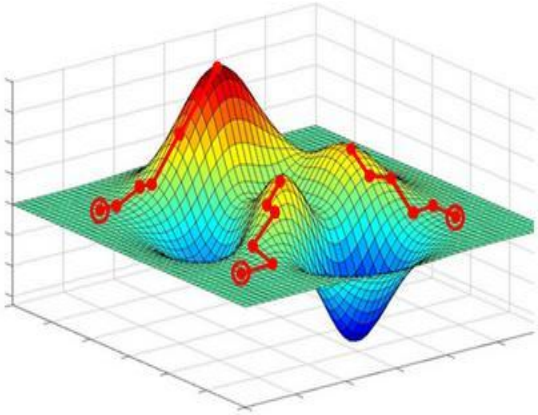
Machine Learning - Linear regression

Gradient descent, minimize cost function.



Machine Learning - Linear regression

Gradient descent, minimize cost function.



Gradient descent algorithm

repeat until convergence {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

(for $j = 1$ and $j = 0$)

}

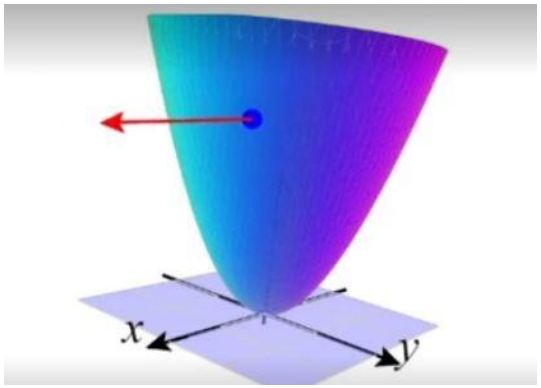
Linear Regression Model

$$h_\theta(x) = \theta_0 + \theta_1 x$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2$$

Machine Learning – The Gradient.

Gradient descent, minimize cost function.



The gradient in each point, points in the direction of steepest ascent.

Gradient descent algorithm

repeat until convergence {

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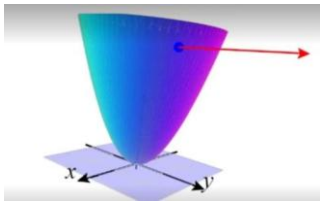
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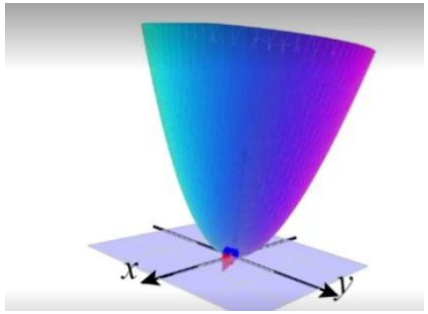
The gradient.

the gradient gives us a direction of steepest ascent. In our gradient descent algorithm, we aim to minimize the value of loss function, therefore at any given point, we need to move in direction where the value of the function decreases the most, i.e. in the opposite direction of gradient which is direction of steepest descent.

Machine Learning – The Gradient.

Gradient descent, minimize cost function.





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$$(\text{for } j = 1 \text{ and } j = 0)$$

}

Linear Regression Model

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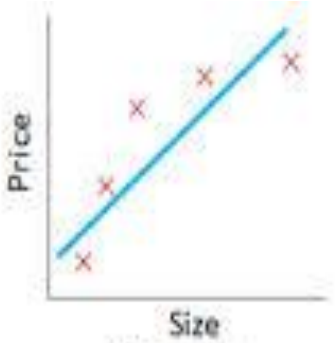
Machine Learning – Linear regression

Gradient descent, minimize cost function.

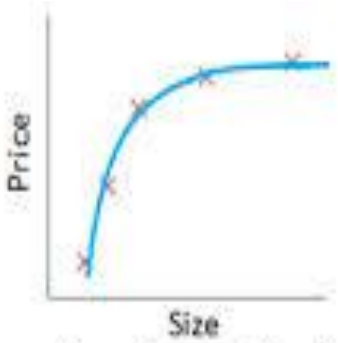
$$
\begin{aligned}
&\text{repeat until convergence } \{ \\
&\quad \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) \\
&\quad \theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)} \\
&\}
\end{aligned}
$$

Machine Learning – The right model.

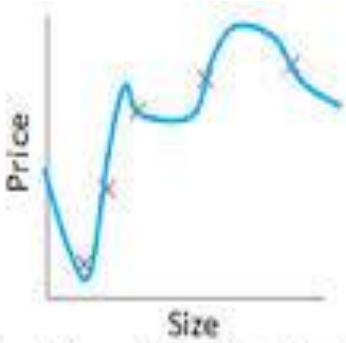
Using the right model: Underfitting, overfitting.



$$\theta_0 + \theta_1 x$$



$$\theta_0 + \theta_1 x + \theta_2 x^2$$



$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$



Machine Learning - Regularization.

Constraining a model to make it simpler and reduce the risk of overfitting is called regularization.

p. 29. v2. ”Hands-On Machine Learning” by Aurelien Geron.

Machine Learning – Feature scaling.

ML doesn't perform well when the input have very different scales.

E.g. use Min-Max scaling (many call this normalization),
so values are shifted and rescaled so they end up ranging
from 0 to 1 (Subtract min value, and divide by max – min).

p. 72. v2. ”Hands-On Machine Learning” by Aurelien Geron.

Exercises.

Today it is all about linear regression.

Lets take a look :)

Exercises 1.2 – 1.7.

Homework

Do the exercises for this week.
Read:
"Hands-On Machine Learning" by Aurelien Geron.
 Chapter 1.
    The Machine Learning Landscape. p. 3 – 37 (p. 3 - 33. v2).
Chapter 4.
    Training models. P. 131 – 151
(p. 113 – 132, v2. p 105 – 126 in 1st edition of the book).

A slightly different (easier?) presentation can be found here:

"Real-World Machine Learning" by Brink et al.
Chapter 1 and 2. (What is Machine Learning, Real world Data).
+
P. 69 – 73 Linear Regression
P. 78 – 84 Overfitting, underfitting, cross validation.